AI: Classical Planning

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Planning Blocks World Problem

Introduction to the Block World Problem

Introduction

Artificial Intelligence (AI) involves rational behavior, meaning that planning—formulating a course of action to accomplish specific goals—is fundamental in AI.

Planning uses a factored representation where the state of the world is described by a set of variables. Actions are expressed in PDDL (Planning Domain Definition Language).

The Blocks World Problem

The Block World Problem is a classic example in AI. It involves a set of blocks, a table, and a robot that can move blocks to achieve a specific configuration.

The robot can perform actions like picking up, putting down, stacking, and unstacking blocks.

Initial State

In this example, the initial configuration is:

- Block A is on top of block C, which is on the table.
- Block B is alone on the table.

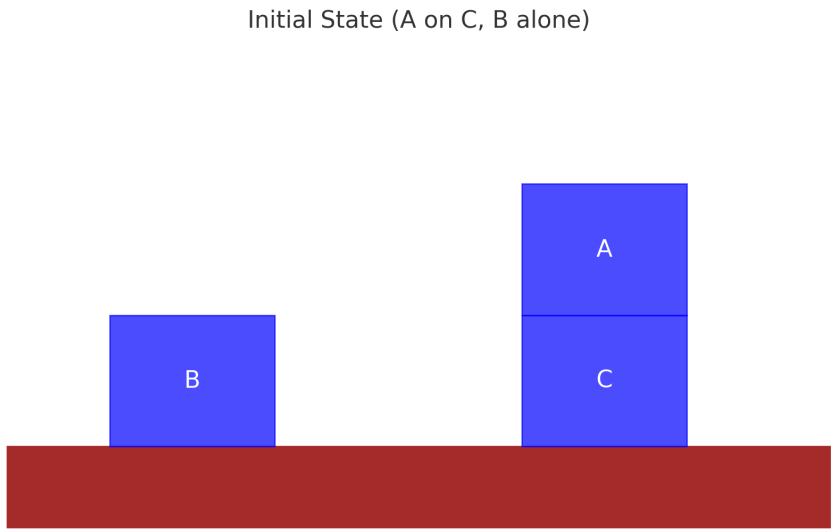


Figure: Initial State: A on C, B alone on the table.

Goal State

The goal configuration is:

- Block C is on the table.
- Block B is on top of block C.
- Block A is on top of block B.

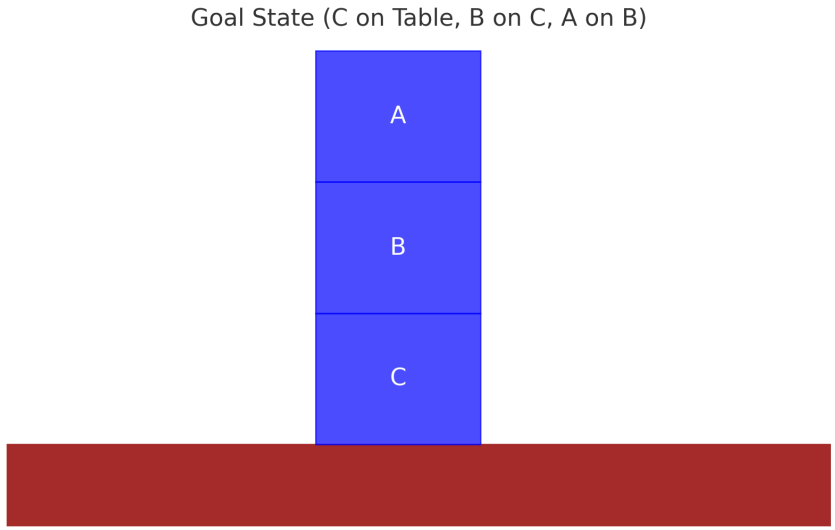


Figure: Goal State: A on B, B on C, C on the table.

Step 1: Unstack A from C

Action: The robot picks up block A from block C.
Result: Block A is in the robot's hand, and block C is clear.

Table: C B (A in hand)

Step 2: Put A on the table

Action: The robot places block A on the table.
Result: Block A is on the table, and the robot's hand is empty.

Table: A B C

Step 3: Pick up B from the table

Action: The robot picks up block B from the table.
Result: Block B is in the robot's hand.

Table: A C (B in hand)

Step 4: Stack B on C

Action: The robot places block B on top of block C.
Result: Block B is on C, and the robot's hand is empty.

Table: A C

B (on C)

Step 5: Pick up A from the table

Action: The robot picks up block A from the table.
Result: Block A is in the robot's hand.

Table: C

B (on C) (A in hand)

Step 6: Stack A on B

Action: The robot places block A on top of block B.
Result: Block A is on B, and the robot's hand is empty.

Table: C

B (on C)

A (on B)

PDDL Representation

Initial State:

$$\begin{aligned}\text{Init}(&\text{On(A, Table)} \wedge \text{On(B, Table)} \wedge \text{On(C, A)} \\ &\wedge \text{Block(A)} \wedge \text{Block(B)} \wedge \text{Block(C)} \\ &\wedge \text{Clear(B)} \wedge \text{Clear(C)})\end{aligned}$$

Goal State:

$$\text{Goal}(\text{On(A, B)} \wedge \text{On(B, C)})$$

PDDL Representation

Actions:

Move Action:

$$\text{Action}(\text{Move}(\text{b, x, y}),$$

PRECONDITION:

$$\text{On}(\text{b, x}) \wedge \text{Clear}(\text{b}) \wedge \text{Clear}(\text{y}) \wedge \text{Block}(\text{b}) \wedge \text{Block}(\text{y})$$
$$\wedge (b \neq x) \wedge (b \neq y) \wedge (x \neq y)$$

EFFECT:

$$\text{On}(\text{b, y}) \wedge \text{Clear}(\text{x}) \wedge \neg\text{On}(\text{b, x}) \wedge \neg\text{Clear}(\text{y})$$

PDDL Representation

Actions:

MoveToTable Action:

$$\text{Action(MoveToTable(b, x),}$$

PRECONDITION:

$$\text{On(b, x)} \wedge \ \text{Clear(b)} \wedge \ \text{Block(b)} \wedge (b \neq x)$$

EFFECT:

$$\text{On(b, Table)} \wedge \ \text{Clear(x)} \wedge \neg\text{On(b, x)}$$

Conclusion

In the Block World Problem, starting from the initial state, the robot rearranges the blocks into the goal state by performing six actions.

This example demonstrates key classical planning techniques, such as sequence-based action planning and reasoning about preconditions and effects.

Air Cargo Transportation Problem

Problem Description: The Air Cargo Transportation Problem involves transporting cargo between multiple airports using airplanes. The goal is to move cargo from its initial locations to designated destinations, considering plane availability and constraints on loading, unloading, and flying.

Example Setup

In this example:

- *Two airports*: $A1$ and $A2$.
- *Two airplanes*: $P1$ and $P2$.
- *Two cargo items*: $C1$ and $C2$.

Initial State

Initial State:

- *Cargo* $C1$ is at airport $A1$.
- *Cargo* $C2$ is at airport $A2$.
- *Airplane* $P1$ is at airport $A1$.
- *Airplane* $P2$ is at airport $A2$.

At(C1, A1), At(C2, A2), At(P1, A1), At(P2, A2)

Goal State

The goal is:

- *Cargo* $C1$ is at airport $A2$.
- *Cargo* $C2$ is at airport $A1$.

$$\text{At(C1, A2)}, \quad \text{At(C2, A1)}$$

Actions

The following actions are available:

- *Load(c, p, a)*: Load cargo c onto airplane p at airport a.
- *Unload(c, p, a)*: Unload cargo c from airplane p at airport a.
- *Fly(p, from, to)*: Fly airplane p from airport *from* to airport *to*.

Air Cargo Transportation Problem: PDDL Representation

Initial State:

$$\begin{aligned}\text{Init}\big(&\text{At}(\text{C1}, \text{SFO}) \land \text{At}(\text{C2}, \text{JFK}) \land \text{At}(\text{P1}, \text{SFO}) \land \text{At}(\text{P2}, \text{JFK}) \\ &\land \text{Cargo}(\text{C1}) \land \text{Cargo}(\text{C2}) \land \text{Plane}(\text{P1}) \land \text{Plane}(\text{P2}) \\ &\land \text{Airport}(\text{JFK}) \land \text{Airport}(\text{SFO})\big)\end{aligned}$$

Goal State:

$$\text{Goal}\big(\text{At}(\text{C1}, \text{JFK}) \land \text{At}(\text{C2}, \text{SFO})\big)$$

Air Cargo Transportation Problem: PDDL Representation

Actions:

- **Action**(Load(c, p, a)):

  PRECONDITION: $At(c, a) \land At(p, a) \land Cargo(c) \land Plane(p) \land Airport(a)$

  EFFECT: $\neg At(c, a) \land In(c, p)$

- **Action**(Unload(c, p, a)):

  PRECONDITION: $In(c, p) \land At(p, a) \land Cargo(c) \land Plane(p) \land Airport(a)$

  EFFECT: $At(c, a) \land \neg In(c, p)$

- **Action**(Fly(p, from, to)):

  PRECONDITION: $At(p, from) \land Plane(p) \land Airport(from) \land Airport(to)$

  EFFECT: $\neg At(p, from) \land At(p, to)$

Air Cargo Transportation Problem: PDDL Representation

Action Sequence: To achieve the goal, apply the following sequence:

$$[\text{Load}(C1, P1, \text{SFO}), \text{Fly}(P1, \text{SFO}, \text{JFK}), \text{Unload}(C1, P1, \text{JFK}),$$
$$\text{Load}(C2, P2, \text{JFK}), \text{Fly}(P2, \text{JFK}, \text{SFO}), \text{Unload}(C2, P2, \text{SFO})]$$

Conclusion

This example illustrates a sequence of actions to achieve the Air Cargo Transportation goal configuration by using actions such as *load*, *unload*, and *fly*.